

Communication-Avoiding Krylov Exponential Integration for Multi-GPU Option Pricing

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Trinity College Dublin

Coláiste na Tríonóide, Baile Átha Cliath

The University of Dublin



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the operator, and one exponential action per step
- 2 **Where the time goes**
profiling, the two CA mechanisms, and a null result



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profiling, the two CA mechanisms, and a null result
- 3 ★ **The regime map**
two a-priori coordinates that predict the opportunity
- 4 ★ **Does it pay?**
a distributed CA integrator on four GPUs

Method in one line

Predict the opportunity *before* building the kernel, then measure whether the prediction holds.

★ marks my own contribution.

1. The problem and the method
2. Where the time goes
3. The regime map
4. Does it pay?



Three-asset Black–Scholes in log prices, backward in $\tau = T - t$, for a **basket call** and a **rainbow min-call**:

$$\frac{\partial u}{\partial \tau} = \underbrace{\frac{1}{2} \sum_{d,d'} \rho_{dd'} \sigma_d \sigma_{d'} \frac{\partial^2 u}{\partial x_d \partial x_{d'}}}_{\text{diffusion+correlation}} + \underbrace{\sum_d \left(r - \frac{1}{2} \sigma_d^2 \right) \frac{\partial u}{\partial x_d}}_{\text{convection}} \underbrace{- ru}_{\text{reaction}} + b(\tau) \quad (1)$$



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Second-order central differences on a uniform grid, $N = n^3$ unknowns:

- A carries **19 nonzeros per row** and $O(N_1 N_2)$ bandwidth, which rules out banded direct solvers at scale.
- Convection is skew-symmetric, so A is **non-symmetric and non-normal**.
- Stiffness grows as $\|A\| \sim O(n^2)$.

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Production resolution is $n = 61$: **226,981 unknowns**, 4.23 M nonzeros.
Every timing in this talk is for the **basket call**, the more expensive of the two.



Discretizing space only leaves time continuous: the PDE becomes a large linear system of ODEs, one equation per grid point.

$$\frac{du}{d\tau} = Au + Bs(\tau), \quad u(0) = u_0 \quad (\text{the payoff}). \quad (2)$$



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For the basket, the payoff and the far-field boundary forcing are

$$u_0(x) = \max\left(\sum_d w_d e^{x_d} - K, 0\right), \quad s(\tau) = \left[\frac{\tau^2}{2}, \tau, 1\right]^T.$$



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The forcing is polynomial, so $s' = Ks$ with K nilpotent. Stacking the two states absorbs it *exactly*, and (2) loses its forcing term altogether:

$$\tilde{A} = \begin{bmatrix} A & B \\ 0 & K \end{bmatrix} \in \mathbb{R}^{(N+3) \times (N+3)}, \quad \tilde{v} = \begin{bmatrix} u \\ s \end{bmatrix} \implies \frac{d\tilde{v}}{d\tau} = \tilde{A}\tilde{v}. \quad (3)$$



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One step from τ_k is then *one matrix-exponential action* on $\tilde{v}_k$, approximated in the Krylov subspace $\mathcal{K}_m(G, \tilde{v}_k)$, $G = h\tilde{A}$:

$$u_{k+1} = e^{G\tilde{v}_k} \approx \beta V_m e^{H_m} e_1, \quad \beta = \|\tilde{v}_k\|. \quad (4)$$

Call this solver **KSM-EI**: a **K**rylov **S**ubspace **m**ethod used as an **e**xponential **i**ntegrator.

1. The problem and the method
2. Where the time goes
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Kernel	$n=31$	$n=61$
SpMV	83.5%	78.0%
MGS	11.4%	16.6%
dense $\exp(H_m)$	0.40%	0.23%
mean m	7.60	8.91

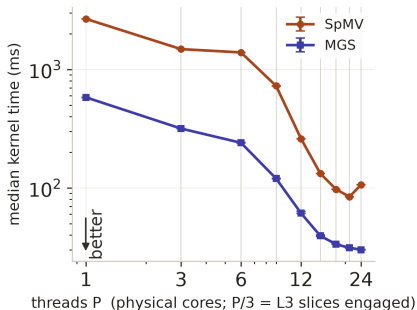
share of KSM-EI solve time

Grids follow Niesen & Wright's option-pricing study, so the semi-discrete ODE error stays comparable with theirs.

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Strong scaling at $n=61$: fixed work, more cores.

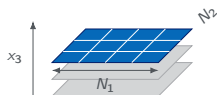
SpMV plateaus until the working set fits the engaged cache. Modified Gram-Schmidt (MGS) keeps improving, but every inner product it performs is a global reduction, and its share grows with m .



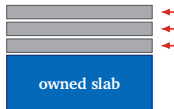
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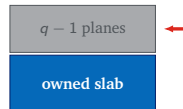
1. Vertical: one deep exchange, not $q - 1$.



one plane:
an $N_1 \times N_2$ cut



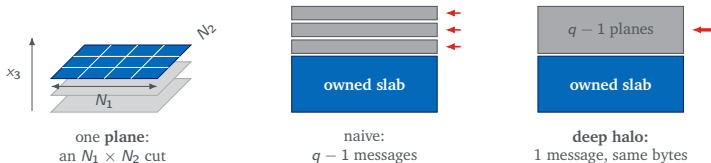
naive:
 $q - 1$ messages



deep halo:
1 message, same bytes

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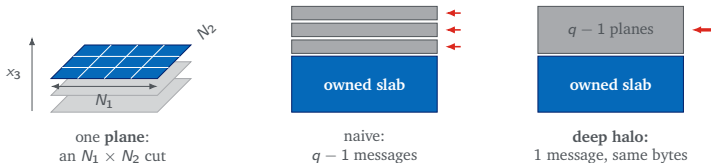


2. Horizontal: one block, not m vectors. Orthogonalize a whole Krylov block at once with **CholeskyQR2**: form the Gram matrix $Z^T Z$, Cholesky-factor it, repeat once for stability.

$$Z_q = [v, Av, \dots, A^{q-1}v] \quad \underbrace{1 + \frac{m(m+3)}{2}}_{\text{MGS}} \longrightarrow \underbrace{1 + 2\lceil m/q \rceil}_{\text{one block}}$$

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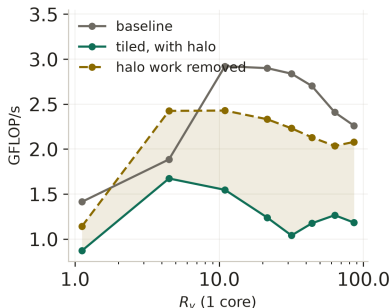


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Messages fall from $q - 1$ to one and reductions by **11–16** $\times$ at the measured $m = 8\text{--}11$, $q = 8$; the price is recomputing the ghost region.

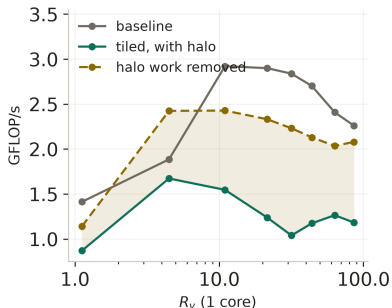
Both mechanisms built and timed on *puffin*, a 24-core Threadripper 3960X on a single NUMA socket.



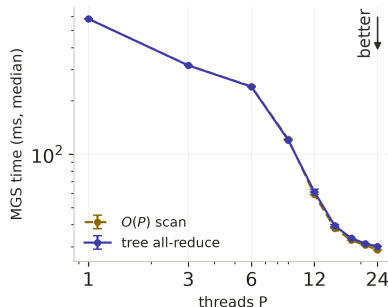
Vertical (L2): the gap is the halo.

The L3 sweep agrees.

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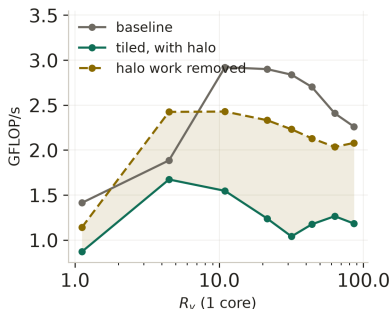


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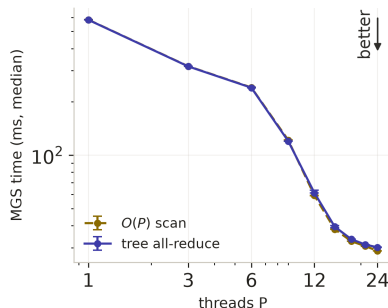


Horizontal: a $1.7\times$ faster all-reduce
leaves MGS within 5%.

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Horizontal: a $1.7\times$ faster all-reduce
leaves MGS within 5%.

11–16 $\times$ fewer reductions, 4–8 $\times$ less modeled traffic.
The saving is real; the speedup is not. So when *does* it pay?

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$$R_v = \frac{\text{Arnoldi working set}}{\text{last-level cache}}$$

The regime map: predict first, then measure



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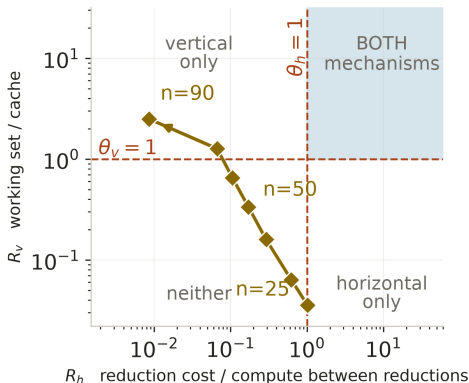
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Eight production grids, $n = 25 \rightarrow 90$, real basket operator, puffin's calibrated constants.

On one socket $R_v R_h$ stays near 0.04 for every P , so the corner is not merely unreachable here: it is unreachable on this topology.



R_v , **from the problem.** Working-set bytes implied by N , the sparsity pattern and the measured Krylov dimension m , over the last-level cache actually available to the team:

$$R_v = \frac{\text{bytes}(N, \text{nnz}, m)}{C_{\text{LLC}}(P)}.$$



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W_{local} , compute between reductions; t_{intra} , t_{cross} , cost of one tree level inside and across a cache domain; L_{in} , L_x , how many levels of each kind the tree crosses; L^* , fitted depth beyond which concurrency hides the cost.



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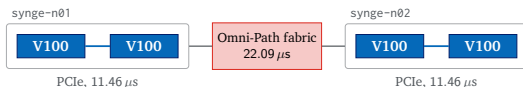
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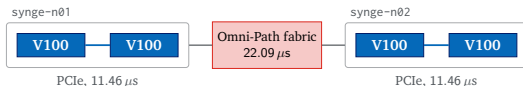
Raising the cost of a reduction is what moves an operator right, and that is a property of topology.

More participants push the operator into the corner

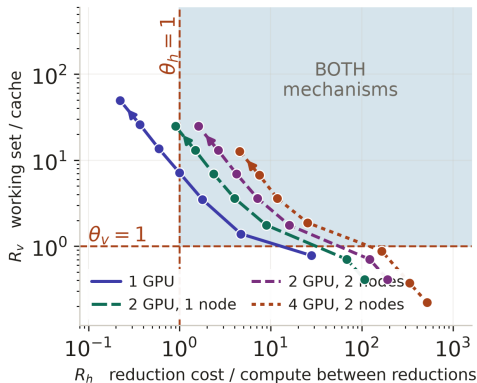


A reduction inside a card costs tens of nanoseconds; one that leaves the node costs **22 μ s**.

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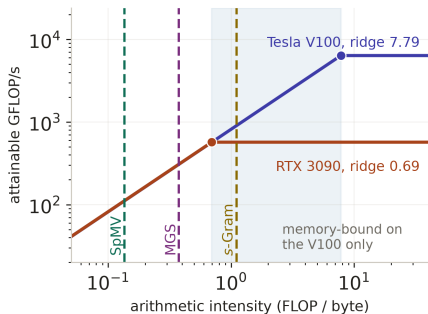


Every added participant raises t_{red} , moving the whole trajectory **right**.

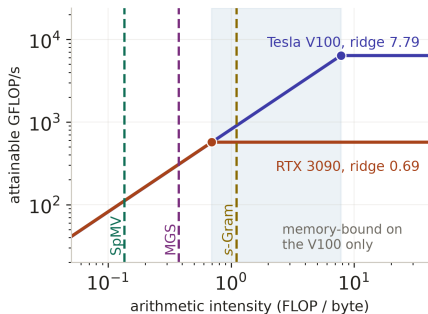
At $n=61$, $m=12$, $q=8$:
 $R_v = 7.2$, $R_h = 1.7$, inside the corner on both cards.

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Reaching the corner is not enough: the device's roofline



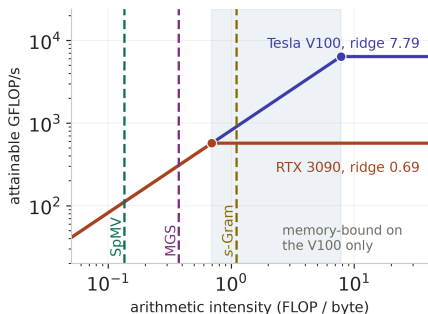
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Nsight Compute, *s*-Gram kernel

% of roof	V100	3090
DRAM	86%	38%
compute	16%	94%
verdict	memory	compute

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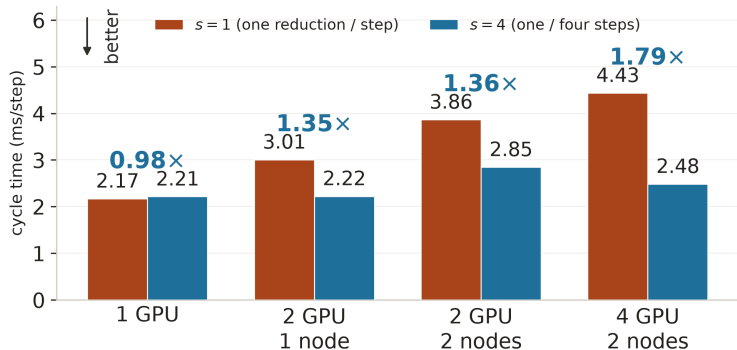
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The memory-bound check

A precondition, before the map is read: compare each kernel's arithmetic intensity (AI , FLOP per byte) with the FP64 ridge, peak FP64 over achieved bandwidth (BW).

$$AI_{\text{kernel}} < \underbrace{\frac{\text{peak FP64}}{BW_{\text{achieved}}}}_{\text{FP64 ridge}} \implies \text{memory-bound, and on the map.}$$

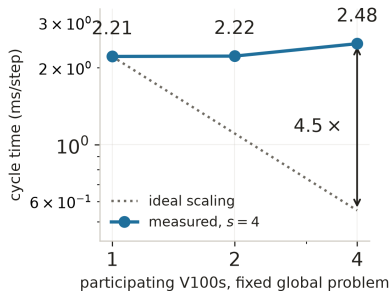
Does communication avoidance pay?



Basket, production grid $n=61$, 100 steps, *exact-depth* halo: both widths do equal numerical work.

Parity on one GPU, and **1.78–1.80×** on four V100s across two nodes.

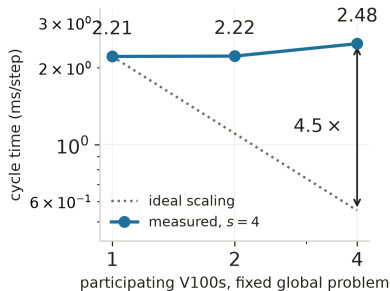
The honest limit: CA helps, distribution still costs



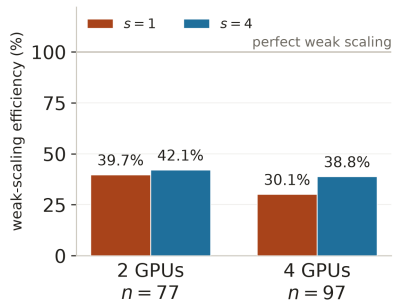
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0.89×, about 22% efficiency.

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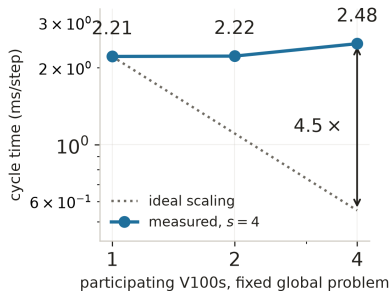


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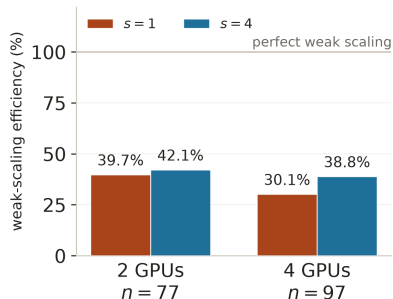


Weak: $\approx 227,000$ rows per GPU held fixed.

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Strong: fixed problem, more GPUs.
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Weak: $\approx 227,000$ rows per GPU held fixed.

At $n=61$ each of four slabs owns only $\approx 57,000$ rows: too little to saturate a V100. Four GPUs earn their place **above** $n = 337$, where the problem stops fitting on one card.



Communication avoidance pays only when the **operator**, the **kernel** and the **machine** all line up.



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What the work establishes

- Two a-priori coordinates that say *which* mechanism is worth measuring, without timing the solver.
- One system where the corner is unreachable and one where more participants reach it, on the same definitions.
- A distributed CA integrator, **1.78–1.80**× faster on four V100s, validated against independent references and **5.8–6.4**× faster than smoothed ADI at equal declared accuracy.



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Future work

- A **tile-aware** R_v predicting read redundancy from tile dimensions, stencil radius and streamed height.
- **Mixed precision** recurrence with higher-precision Gram accumulation. Not free: it can cost numerical stability, so the first job is identifying which parts of the solve are sensitive to the change.
- **H100-class devices and faster links**: does the crossover transfer beyond Synge?



This work stands on a great deal of help.

My thanks to my supervisor, to the staff and students of the MSc in High-Performance Computing, to everyone who kept the clusters running, and to my family, for the teaching, the patience and the support that made it possible.

“If I have seen further it is by standing on the shoulders of Giants.”

Isaac Newton, letter to Robert Hooke, 5 February 1675

Questions?

Code: github.com/kevinknights29/caksm

Thesis: [.../caksm/docs/thesis/main.pdf](https://github.com/kevinknights29/caksm/blob/main/docs/thesis/main.pdf)

Appendix



Are the prices still right, and how does the solver compare with an established method?

KSM-EI is the Krylov exponential integrator.

ADI-HV-S the smoothed Hundsdorfer–Verwer ADI scheme.

Equal declared accuracy, one thread, $n=61$

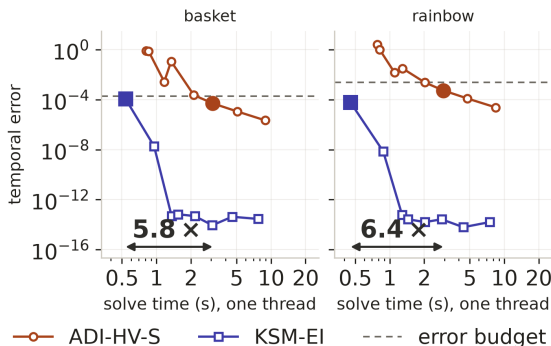
Where the price error actually is

Payoff	Method	Steps	Median (s)
Basket	ADI-HV-S	50	3.090
Basket	KSM-EI	1	0.533 5.8×
Rainbow	ADI-HV-S	50	2.936
Rainbow	KSM-EI	1	0.460 6.4×

Source of Δ	Basket	Rainbow
reference	6.8×10^{-6}	$< 10^{-13}$
domain	1.0×10^{-4}	4.5×10^{-6}
spatial	2.9×10^{-3}	3.6×10^{-2}
ADI temporal	2.2×10^{-6}	2.4×10^{-5}
KSM temporal	5.3×10^{-15}	3.6×10^{-15}

Both solvers refine at second order against independent references.

Basket Δ errors are 0.011–0.020%. The ADI comparison is **Dang, Christara & Jackson (2010)**, my last seminar paper.



One thread, $n=61$; each marker is one step count, 1 step at the left to 400 at the right.

ADI-HV-S needs **50 steps** to meet the error budget. KSM-EI reaches machine precision in **one**.

ADI-HV-S is second order in time, so accuracy has to be bought with steps. KSM-EI's curve is vertical: past the first step there is no temporal error left to remove, and further steps buy only time.



A monomial block $[v, Av, \dots, A^{q-1}v]$ loses conditioning fast, so CholeskyQR2 needs a guard.

In practice. The solver narrows the block whenever

$$\kappa_2(Z_q) > u^{-1/2} \approx 9.5 \times 10^7.$$

In theory. CholeskyQR2's sufficient condition is dimension-dependent,

$\delta = 8\kappa_2(Z_q)\sqrt{(N+3)qu + q(q+1)u} \leq 1$,
which allows only $\kappa_2 \lesssim 1.25 \times 10^4$ at our size. The measured block reaches 6.0×10^6 , so $\delta \approx 482$: the theorem is **inconclusive here, not violated**.

So the claim rests on the guard, a successful factorization, the orthogonality measured on the right, and referee agreement.

Orthogonality loss, $n=61, m=8, q=4$

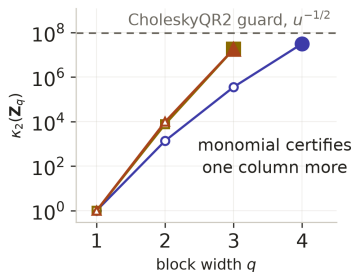
Method	$\ I - Q^T Q\ $	Time
CholeskyQR2	4.0×10^{-14}	1.00×
TSQR	3.5×10^{-14}	3.06×
Block GS, reorth.	6.5×10^{-14}	1.24×

lower is better in both columns

Width admitted by the guard

Basis	$n=31$	$n=61$
Monomial	4	4
Newton (Leja)	3	3
Chebyshev	4	3

higher is better: a wider block means fewer reductions



—○— monomial —□— Newton —△— Chebyshev

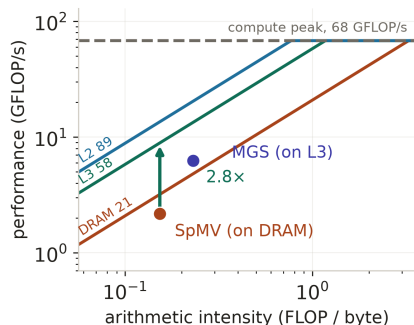
Basket, $n=61$. Each line stops at the widest block that basis certifies (filled marker).

m stays small. The endpoint defect is met at a mean $m \approx 8$, so the recurrence never needs $q > 4$.

Small q is the safe regime. Monomial conditioning is only a problem for wide blocks. At $q \leq 4$ it stays under the guard, and grows *more slowly* than either alternative.

The alternatives do not pay. Newton and Chebyshev exist to hold conditioning down at large q . Here they are worse at every width, certify one column fewer, and need spectral estimates the monomial does not.

Stability here comes from m being small, not from a clever basis.



One core, $n=61$.

$$\text{gain} = \frac{\min(BW_{\text{fast}} \cdot AI, \text{peak})}{\min(BW_{\text{slow}} \cdot AI, \text{peak})} = \frac{58.2}{20.9} = 2.8\times \quad \text{at } AI = 0.15.$$

Kernel

AI

Tier

SpMV

0.15

DRAM

MGS

0.23

L3

dense $\exp(H_m)$

1.76

L2

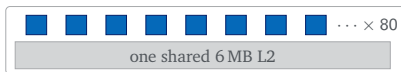
arithmetic intensity, FLOP per byte



CPU: puffin, Threadripper 3960X



GPU: synge, Tesla V100



CPU

- Cache **grows with the team**: each engaged core complex brings its own L3 slice.
- R_v scales with the engaged core count P , so the two axes are coupled and $R_v R_h$ is roughly P -independent.
- Reductions cross **2** levels.

GPU

- Cache is **fixed**: engaging more SMs adds none.
- R_v stops depending on how many SMs are engaged, so **grid size alone moves it**.
- Reductions cross **5** levels, one of them a kernel launch. Coalescing also makes the access pattern far more sensitive.

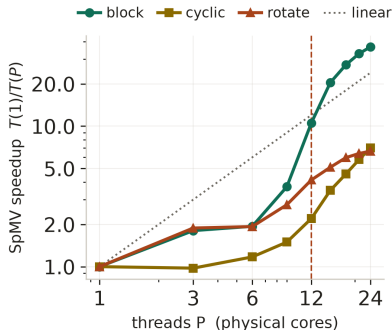


$$R_v R_h = g(m) R(m) \Lambda, \quad \Lambda = \frac{t_{\text{red}} \cdot BW}{C} \implies t_{\text{red}}^* = 0.69 \mu\text{s}$$

priced at the production point $n=61, m=17$.

Reduction level	$t_{\text{red}} (\mu\text{s})$	Corner
warp shuffle	0.42	closed
thread block	0.79	open
grid, 2nd launch	5.54	open
device, PCIe	11.46	open
node, fabric	22.09	open

Puffin, for comparison: $t_{\text{intra}} = 120.8 \text{ ns}$, $t_{\text{cross}} = 652.8 \text{ ns}$, $L^* = 2.20$, $R^2 = 0.98$.



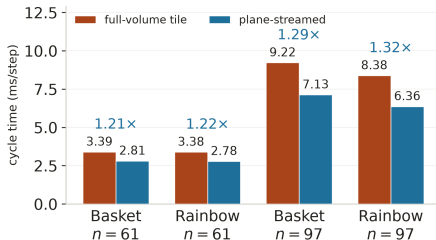
Same operator and byte count;
only the row-to-core map changes.

schedule	Rows per core	Speedup
block	contiguous band	36.7×
cyclic	same bytes, dealt out	7.0×
rotate	contiguous, reshuffled	6.6×

the OpenMP schedule clause

At the knee the implied 176 GB/s exceeds puffin's ~ 95 GB/s DRAM roof, so it comes from L3. **Contiguity, not volume, keeps a slice resident.**

On one NUMA node no reduction costs more than a core-complex hop, so synchronization is never dominant: the lever here is **access order**, not communication avoidance.



One V100. The x - y tile is fixed at 16×8 ; only the order in which a block walks z changes.

Nsight counters, full-volume $\rightarrow$ plane-streamed

	$n=61$	$n=97$
thread blocks launched	2,048 $\rightarrow$ 256	8,125 $\rightarrow$ 637
executed instructions	2.9×	3.4×
global-load sectors	2.6–4.2×	

Each plane is fetched from global memory *once* and reused down the segment.

This does not replace s -step; it composes with it.



FP64 footprint	$n=61$	$n=337$
unknowns N (millions)	0.23	38.3
one state vector (MB)	1.8	306
Krylov basis, $m=39$ (GB)	0.07	11.9
share of a 16 GB V100	0.4%	75%

At $n = 337$ the solve converges but is still not accepted

Maximum $m = 32$, residual 9.85×10^{-9} : the *convergence* gate passes. The *boundary-state* gate fails, at 6.40×10^{-5} , and the cause is scaling rather than convergence. The fixed augmentation leaves $\|B\|_1 = 4.15 \times 10^{10}$, so $\tilde{A}$ is badly scaled and the three-component forcing tail loses its digits. A power-of-two similarity rescaling, $\eta = 2^{-36}$, brings $\|B\|_1$ to 0.603 and the tail error to 4.41×10^{-13} , and the run then passes.

The original stop remains the reported result.



Gate	What it checks	Tolerance
conditioning	$\kappa_2(Z_q)$ before every factorization	$u^{-1/2} \approx 9.5 \times 10^7$
convergence	endpoint defect η_m , at every step	10^{-8}
boundary state	the forcing tail against its closed form	10^{-8}
single-GPU	distributed state against a one-GPU run	$10^{-13} \ \text{state}\ $
referee	against an Al-Mohy–Higham solve	stated per arm

A result is reported only if **all five** pass. Each is a different quantity, so tightening one does not repair another: the $n = 337$ probe converges and still fails at *boundary state*.